Can Anything Beat VIX? A Systematic Out-of-Sample Evaluation of Eleven Signal Families for Equity Volatility Forecasting and Volatility Timing*

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Abstract

We conduct a systematic out-of-sample horse race among eleven pre-specified signal families—cross-asset volatility momentum, VIX term structure, behavioral sentiment, the variance risk premium, multi-asset portfolio optimization, equal risk contribution, Bitcoin volatility, the yield curve slope, Google Trends fear proxies, overnight VIX changes, and calendar anomalies—evaluating whether any can improve upon VIX alone for equity volatility forecasting and volatility-timing portfolio construction. Using daily S&P 500 data spanning 1993–2026, we apply a unified forecast pipeline with strict lag enforcement, transaction-cost accounting, and the Harvey et al. (2016) multiple-testing threshold ($|t| > 3.0$). The main finding is strongly negative: not a single signal family produces a statistically significant out-of-sample improvement, whether measured by incremental R^2 for volatility forecasting or by Sharpe ratio gains for volatility-timing strategies. The VIX–realized volatility R^2 ranges from 0.24 to 0.64 across five non-overlapping eras spanning 33 years (coefficient of variation = 0.33), demonstrating time-invariance. We frame VIX sufficiency as a rational outcome of option-market information aggregation and show that volatility timing functions as drawdown insurance—costly

*All errors are my own. Data and replication code are available upon request.

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in Sharpe terms (average drag 3.49%/year for 12/VIX) but welfare-improving for investors with CRRA risk aversion $\gamma \geq 4.5$. The informative null sharpens the research frontier: future gains require either higher-frequency realized measures or genuinely exogenous information not already embedded in option prices.

Keywords: VIX, volatility forecasting, volatility timing, out-of-sample evaluation, variance risk premium, multiple testing, option-implied volatility, drawdown insurance

JEL Classification: C53, G11, G12, G17

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1 Introduction

The Chicago Board Options Exchange Volatility Index (VIX) is the single most widely used measure of expected equity market volatility. Derived from the prices of S&P 500 index options, VIX represents the market's consensus forecast of 30-day annualized volatility and has been labeled the "fear gauge" of financial markets (Whaley, 2000). A vast literature documents that VIX is a powerful, though biased, predictor of future realized volatility (Christensen and Prabhala, 1998; Jiang and Tian, 2005; Bekaert and Hoerova, 2014). Yet the practical question facing portfolio managers, risk officers, and retail investors remains open: *can any observable signal improve upon VIX for forecasting equity volatility and timing equity exposure?*

This question is not merely academic. Volatility-timing (VT) strategies, which scale equity exposure inversely to expected volatility, have attracted substantial interest since the seminal work of Fleming et al. (2001) and the volatility-managed portfolio framework of Moreira and Muir (2017). If a signal can forecast volatility more accurately than VIX, it can potentially improve the risk-return trade-off of these strategies. Conversely, if VIX already aggregates all publicly available information relevant to near-term volatility, then the search for a "better VIX" is futile, and research efforts should be redirected.

The existing literature offers fragments of evidence but no systematic verdict. Individual studies have examined specific signal categories—the variance risk premium (Bollerslev et al., 2009), cross-asset spillovers (Diebold and Yilmaz, 2012), economic indicators (Schwert, 1989), and sentiment measures (Baker and Wurgler, 2006)—but each study uses its own sample period, evaluation metric, and benchmark specification, making cross-study comparison unreliable. Moreover, most studies do not enforce the multiple-testing corrections that Harvey et al. (2016) argue are essential when many predictors are examined.

We address this gap by conducting the first comprehensive, pre-registered horse race among eleven signal families, evaluated under a unified framework with five methodological safeguards:

(i) **Pre-specification**: All eleven signal families and their construction rules are defined before examining out-of-sample results, eliminating data-snooping in signal selection.

(ii) **Strict lag enforcement**: Every signal is lagged one trading day (`signal.shift(1)`) before being applied to returns, preventing look-ahead bias—a surprisingly common error that we document has inflated Sharpe ratios by up to 373% in prior work.[1]

(iii) **Transaction costs**: All strategies deduct 5 basis points per leg on each weight change, applied to both buy and sell sides.

(iv) **Multiple-testing control**: We adopt the Harvey et al. (2016) threshold of $|t| > 3.0$ for all Diebold-Mariano tests, accounting for the implicit multiplicity of testing eleven signal families.

(v) **Cross-era stability**: Results are evaluated separately across five non-overlapping eras spanning 1993–2026, preventing any single regime from driving conclusions.

Our main finding is strongly negative but highly informative. Across 11 signal families, 36 VIX sufficiency tests, and five market eras totaling 8,325 trading days, we find:

- No signal family produces a statistically significant out-of-sample R^2 improvement over VIX alone (maximum incremental $R^2 = 0.038$, all Diebold-Mariano $|t| < 3.0$).
- No signal-based strategy statistically outperforms the simple 12/VIX benchmark or the static 50/50 SPY/GLD allocation (maximum Sharpe improvement = +0.010, not significant).
- VIX's forecasting power is time-invariant: the VIX–realized volatility R^2 ranges from 0.24 (low-volatility QE era, 2012–2020) to 0.64 (post-dot-com era, 2000–2007), with a coefficient of variation of only 0.33.

[1] In separate robustness analysis, careful code review identified look-ahead bias in four backtests, one of which produced a Sharpe ratio of 1.68 that collapsed to 0.355 after lag correction.

- Even the most economically motivated signal—the variance risk premium—achieves an in-sample t-statistic of 3.51 for return prediction but delivers out-of-sample $R^2 < 0.3\%$.

We interpret this null through the lens of option-market information aggregation. VIX is not simply a backward-looking statistical estimate; it is the forward-looking consensus of thousands of option traders who collectively incorporate cross-asset signals, macroeconomic data, sentiment, and term structure information into their pricing. The eleven signal families we test are precisely the inputs that option traders use. VIX, by construction, already contains them.

This interpretation leads to our second contribution: a reconceptualization of VT strategies. Rather than viewing VT as an alpha-generating timing strategy, we propose the *drawdown insurance* framework. VT sacrifices average returns (3.49%/year for 12/VIX, 2.12%/year for EWMA) in exchange for maximum drawdown reduction. This trade-off is welfare-improving for investors with CRRA risk aversion $\gamma \geq 4.5$, making VT rational for moderately risk-averse investors even though it underperforms buy-and-hold on standard metrics.

Our third contribution concerns the research frontier. The informative null delineates what *cannot* work (daily signals already in the option-market information set) from what *might* work: genuinely exogenous shocks (e.g., geopolitical events, pandemic declarations) that arrive between option-market pricing updates, and possibly higher-frequency realized measures (5-minute data) whose relative performance is as yet unresolved. A small-sample pilot ($N = 37$) directly comparing daily HAR-ABS (QLIKE = 0.077) with 5-minute HAR-RV (QLIKE = 0.109) found that the daily model *outperformed* the 5-minute model by 41.8% on this limited sample—the reverse of what the high-frequency literature conventionally predicts. This preliminary result is not powered to establish that 5-minute data cannot improve forecasts; it shows only that under this pilot specification and sample, 5-minute HAR-RV did not beat daily HAR-ABS. A larger-scale experiment (planned) will use an extended 5-minute SPY dataset to resolve whether 5-minute information carries incremental predictive content beyond daily closing-price

VIX. We therefore flag the 5-minute frontier as open but not yet empirically established.

The remainder of the paper is organized as follows. Section 2 reviews the theoretical and empirical foundations of VIX as a benchmark. Section 3 describes the data and the eleven signal families. Section 4 details the forecast design and benchmark models. Section 5 explains the statistical evaluation framework. Section 6 describes the volatility-timing strategy design. Section 7 presents the main results and robustness checks. Section 8 discusses why the null is informative. Section 9 concludes.

2 Why VIX is the Benchmark

2.1 Theoretical Foundations

The VIX index, introduced by the CBOE in 1993 and reformulated in 2003, measures the risk-neutral expected variance of the S&P 500 over the next 30 calendar days. Under the model-free framework of Britten-Jones and Neuberger (2000) and Jiang and Tian (2005), VIX squared equals the risk-neutral expectation of integrated variance:

$$\mathrm{VIX}^2 = \frac{2}{\tau}\sum_i \frac{\Delta K_i}{K_i^2} e^{r\tau} Q(K_i) - \frac{1}{\tau}\left(\frac{F}{K_0} - 1\right)^2, \tag{1}$$

where τ is time to expiration, K_i are strike prices, $Q(K_i)$ are midpoint option prices, F is the forward index level, and K_0 is the first strike below F. This construction aggregates information from the entire volatility smile, making VIX a sufficient statistic for the market's assessment of near-term volatility under risk-neutral pricing.

The link between VIX and future realized volatility (RV) is governed by the variance risk premium (VRP):

$$\mathbb{E}^Q[\sigma^2_{t,t+\tau}] = \mathbb{E}^P[\sigma^2_{t,t+\tau}] + \mathrm{VRP}_t, \tag{2}$$

where $\mathbb{E}^Q$ and $\mathbb{E}^P$ denote risk-neutral and physical expectations, respectively. VIX overestimates realized volatility on average (the VRP is positive approximately 85% of days in our sample), but this bias is systematic and does not impair VIX's role as a forecasting benchmark—it merely shifts the intercept of the VIX–RV regression.

2.2 Empirical Evidence for VIX Forecasting Power

A long line of empirical work confirms VIX's superiority over time-series models for volatility forecasting. Christensen and Prabhala (1998) show that VIX subsumes the information in historical volatility for forecasting realized volatility. Jiang and Tian (2005) demonstrate that model-free implied volatility dominates both GARCH and Black-Scholes implied volatility in encompassing regressions. Bekaert and Hoerova (2014) decompose VIX into a conditional variance component and a variance risk premium, finding that the conditional variance component tracks realized volatility closely.

More recently, Bozovic (2024) examines VIX-managed portfolios and finds that VIX-based timing produces economically significant improvements in portfolio performance, though the mechanism operates through drawdown reduction rather than return enhancement—consistent with our insurance framework.

2.3 The Information Aggregation Argument

We argue that VIX's forecasting dominance is not accidental but structural. Options markets aggregate information from heterogeneous traders—fundamental analysts who track earnings volatility, macro strategists who monitor the yield curve, systematic traders who exploit cross-asset correlations, and market makers who process order flow. Each of the eleven signal families we test represents one channel of information that at least some option traders monitor. VIX, as the price-weighted consensus of their collective activity, already incorporates these inputs.

This argument generates a testable prediction: signals that are *already in the option-market information set* should show zero incremental forecasting power after controlling for VIX, while signals that are *outside* this set (e.g., truly private information, intraday microstructure) might still add value. Our empirical results are consistent with this prediction.

2.4 Prior Horse Races

Several studies have compared VIX to specific alternative signals. Corsi (2009) proposes the HAR-RV model using multi-frequency realized volatility components and shows it forecasts well, but does not benchmark against VIX. Bucci (2020) finds that neural networks can improve realized volatility forecasts but does not test whether the improvement survives after conditioning on VIX. Bollerslev et al. (2009) show that VRP predicts equity returns (not volatility), but subsequent out-of-sample tests have been mixed.

Our contribution differs from these studies in three ways: (i) we test eleven signal families simultaneously rather than one at a time; (ii) we use a unified evaluation pipeline with consistent lag, cost, and statistical conventions; and (iii) we frame the null as informative rather than disappointing, connecting it to option-market efficiency.

3 Data and the Eleven Signal Families

3.1 Core Data

Our primary dataset consists of daily closing prices for the S&P 500 index (via SPY ETF), the CBOE Volatility Index (VIX), VIX 3-month futures index (VIX3M), gold (GLD ETF), and the 10-year and 3-month U.S. Treasury yields (TNX, IRX). All price data are sourced from Yahoo Finance. The full sample spans January 1993 to March 2026, yielding 8,325 trading days for the VIX–SPY pair. Individual signal families have shorter samples depending on data availability (detailed in Table ??).

We compute 22-day realized volatility as:

$$RV_t^{(22)} = \sqrt{\frac{252}{22} \sum_{i=0}^{21} r_{t-i}^2}, \tag{3}$$

where $r_t = \ln(P_t/P_{t-1})$ is the daily log return. This serves as the forecast target for all volatility prediction exercises.

3.2 The Eleven Signal Families

We pre-specify eleven signal families spanning six broad categories: cross-asset, derivatives-based, behavioral, macroeconomic, alternative data, and calendar. Table ?? summarizes each family.

Notes: All signals are computed using information available at market close on day $t-1$ and applied t returns. Google Trends data are weekly and converted to daily by forward-filling. Sample sizes vary availability. VIX3M begins in 2008; Bitcoin reliable pricing begins in 2015; Google Trends via pytrends b 2004 but we use 2010+ for stability.

3.2.1 Family 1: Cross-Asset Volatility Momentum

Cross-asset volatility spillovers have been documented by Diebold and Yilmaz (2012) and Barunik et al. (2016). We compute the 5-day change in 22-day realized volatility for five asset classes: bonds (TLT), oil (USO), U.S. dollar (UUP), gold (GLD), and credit (HYG minus LQD spread). A composite momentum signal averages the normalized changes. The hypothesis is that volatility shocks propagate across asset classes with a lead-lag structure.

3.2.2 Family 2: VIX Term Structure

The VIX term structure—the ratio of spot VIX to VIX3M—has been proposed as a regime indicator. Persistent contango (VIX < VIX3M) is associated with complacency, while backwardation signals acute fear (Luo and Zhang, 2019). We test whether this ratio adds information beyond VIX level for both volatility prediction and strategy timing.

3.2.3 Family 3: Behavioral Sentiment Index

Investor sentiment affects asset prices through both rational and behavioral channels (Baker and Wurgler, 2006). We construct a composite Behavioral Sentiment Index (BSI) using four components: the CBOE SKEW index, the VIX/VIX3M ratio, VIX 22-day momentum, and VIX level. Each component is converted to a percentile rank over a

252-day rolling window, and the BSI is the equal-weighted average.

3.2.4 Family 4: Variance Risk Premium

The variance risk premium ($\text{VRP} = \text{VIX} - \text{RV}^{(22)}$) has attracted attention since Bollerslev et al. (2009) showed it predicts equity returns. We distinguish between VRP as a *return* predictor (Bollerslev's original finding) and VRP as a *volatility* predictor, testing whether VRP carries orthogonal information about future volatility after controlling for VIX level.

3.2.5 Family 5: Multi-Asset Portfolio Optimization

DeMiguel et al. (2009) famously showed that $1/N$ portfolios outperform optimized portfolios out of sample. We test four optimization methods (equal-weight, minimum variance, maximum diversification, risk parity) across seven asset subsets (2 to 9 assets including SPY, GLD, TLT, QQQ, EEM, VWO, BND, DBC, IEFA), evaluating whether expanding the asset universe beyond SPY/GLD improves risk-adjusted returns.

3.2.6 Family 6: Equal Risk Contribution

The Equal Risk Contribution (ERC) portfolio (Maillard et al., 2010) equalizes marginal risk contributions through constrained optimization. We implement ERC with monthly hold-and-drift rebalancing and test 2-, 3-, and 4-asset variants against the static 50/50 SPY/GLD benchmark.

3.2.7 Family 7: Bitcoin Volatility

Bitcoin has been proposed as a "digital gold" and potential diversifier (Bouri et al., 2017). We test whether Bitcoin realized volatility contains Granger-causal information for VIX and whether a Bitcoin allocation improves portfolio performance. The *fear channel*—whether crypto stress transmits to traditional markets—is examined through asymmetric Granger causality.

3.2.8 Family 8: Yield Curve Slope

The yield curve slope (10Y − 3M Treasury spread) is a well-known recession predictor (Estrella and Mishkin, 1998). We test whether curve inversion contains volatility information beyond VIX, exploiting the fact that yield curve movements are slow-moving and potentially capture different risk horizons than VIX.

3.2.9 Family 9: Google Trends Fear Proxy

Internet search behavior provides a real-time measure of retail investor attention and fear (Da et al., 2011; Preis et al., 2013). We construct a composite Google Fear Index from the search volume of five terms ("stock market crash," "recession," "VIX," "market crash," "bear market"), using rolling z-scores over a 52-week window.

3.2.10 Family 10: Overnight VIX Changes

Overnight price changes reflect the processing of after-hours news, earnings announcements, and geopolitical events (Bollerslev et al., 2020). We use the absolute overnight VIX change ($|\text{VIX}_{\text{open},t} - \text{VIX}_{\text{close},t-1}|$) as a proxy for the information content of overnight news.

3.2.11 Family 11: Calendar Anomaly

The "Halloween effect" and other calendar anomalies have been documented in equity markets (Bouman and Jacobsen, 2002). We test whether VIX-based timing is simply a disguised calendar strategy by examining the correlation between VIX-implied weights and monthly/seasonal dummies.

4 Forecast Design and Benchmark Models

4.1 Unified Forecast Pipeline

All forecast evaluations follow a unified pipeline to ensure comparability:

1. **Signal construction**: Each signal is computed using only information available at the close of day $t - 1$.

2. **Lag enforcement**: The signal is explicitly shifted by one trading day (`signal = signal.shift(1)`) before any use in forecasting or portfolio construction.

3. **Rolling estimation**: Where applicable, parameters are estimated on a rolling window of 252 trading days (approximately one year), re-estimated daily.

4. **Forecast target**: The dependent variable is 22-day forward realized volatility, $RV_{t+1:t+22}$.

4.2 VIX Benchmark

The VIX benchmark model is a simple linear regression:

$$RV_{t+1:t+22} = \alpha + \beta \cdot \mathrm{VIX}_t + \varepsilon_t. \tag{4}$$

This is our primary benchmark. Any candidate signal must demonstrate statistically significant improvement over this specification.

4.3 Incremental Information Test

For each signal family j, we estimate:

$$RV_{t+1:t+22} = \alpha + \beta_1 \cdot \mathrm{VIX}_t + \beta_2 \cdot \mathrm{Signal}_{j,t} + \varepsilon_t, \tag{5}$$

and test $H_0 : \beta_2 = 0$. We report both the in-sample t-statistic for β_2 and the partial correlation $r_{\text{partial}} = \mathrm{cor}(RV, \mathrm{Signal}_j \mid \mathrm{VIX})$.

4.4 Out-of-Sample R^2

Out-of-sample forecasting accuracy is measured by:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t \in \text{OOS}} (RV_t - \hat{RV}_t^{\text{model}})^2}{\sum_{t \in \text{OOS}} (RV_t - \hat{RV}_t^{\text{VIX}})^2}, \tag{6}$$

where positive values indicate improvement over VIX. Note that this is defined relative to VIX (not the historical mean), making it a stringent test.

5 Statistical Evaluation and Multiple-Testing Control

5.1 Diebold-Mariano Test

We use the Diebold and Mariano (1995) test to compare forecast accuracy, with squared forecast errors as the loss function:

$$d_t = e^2_{t,\text{VIX}} - e^2_{t,\text{model}}, \tag{7}$$

where $e_{t,k}$ denotes the forecast error from model k. The DM statistic is:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/T}}, \tag{8}$$

with Newey-West standard errors to account for serial correlation in multi-step-ahead forecasts.

5.2 Harvey (2016) Multiple-Testing Threshold

Harvey et al. (2016) argue that the conventional $|t| > 1.96$ threshold produces excessive false discoveries when many predictors are tested. They recommend $|t| > 3.0$ as a minimum threshold for asset pricing tests. Given that we test eleven signal families, this threshold is conservative but appropriate. We adopt $|t| > 3.0$ as the significance standard throughout.

5.3 Cross-Era Validation

To prevent any single market regime from driving conclusions, we partition the sample into five non-overlapping eras:

1. **Dot-Com** (1993–2000): Rising equity markets, Asian crisis, LTCM
2. **Post-Dot-Com** (2000–2007): Tech bust, recovery, housing boom
3. **GFC** (2007–2012): Global financial crisis, European debt crisis
4. **Low-Vol QE** (2012–2020): Quantitative easing, low volatility regime
5. **COVID/Inflation** (2020–2026): Pandemic shock, inflation, rate hikes

A finding is considered robust if it holds in at least 4 of 5 eras.

6 Volatility-Timing Strategy Design

6.1 The 12/VIX Strategy

Our primary VT strategy is the 12/VIX rule:

$$w_t^{\mathrm{SPY}} = \min\left(\frac{12}{\mathrm{VIX}_{t-1}}, 1\right), \tag{9}$$

where the equity weight is capped at 100% and the complement $(1 - w_t^{\mathrm{SPY}})$ is allocated to GLD (or cash in pre-GLD eras). This strategy has several desirable properties:

- **Zero parameters**: The numerator 12 approximates long-run average VIX.
- **Smooth weights**: The concave $1/x$ mapping produces gradual weight changes, limiting turnover (mean daily $|\Delta w| = 0.035$, annual turnover $\approx 8.9\times$).

- **Lag-robust**: Because weights change slowly, the 1-day lag between signal and execution has negligible impact.
- **Crowding-resistant**: Market impact analysis shows that even \$10 billion in AUM would reduce Sharpe by less than 60 basis points, and the VIX feedback loop is damping (converging) rather than amplifying.

6.2 Baseline: 50/50 SPY/GLD

The baseline is a static 50/50 allocation to SPY and GLD with monthly rebalancing. This allocation is approximately optimal by construction: SPY annualized volatility (19.3%) ≈ GLD annualized volatility (18.3%), so the 50/50 split closely approximates the risk parity solution (Maillard et al., 2010).

6.3 Transaction Costs

We deduct 5 basis points per leg for each weight change:

$$\mathrm{TX}_t = 0.0005 \times \left(|w_t^{\mathrm{SPY}} - w_{t-1}^{\mathrm{SPY}}| + |w_t^{\mathrm{GLD}} - w_{t-1}^{\mathrm{GLD}}|\right). \tag{10}$$

Net portfolio return is $r_t^{\mathrm{net}} = r_t^{\mathrm{gross}} - \mathrm{TX}_t$.

7 Main Results and Robustness

7.1 Volatility Forecasting: No Signal Beats VIX

Table 2 presents the main results for all eleven signal families. The key finding is unanimous: no signal produces a statistically significant out-of-sample improvement over VIX.

Table 2: Volatility Forecasting: Eleven Signal Families vs. VIX

#	Signal Family	Partial r|VIX	IS ΔR^2	IS t-stat	OOS R^2	DM $\|t\|$
1	Cross-asset vol momentum	0.087	-0.022	-1.45	-0.022	1.45
2	VIX term structure	0.181	0.033	17.6***	0.010	0.87
3	Behavioral sentiment	0.091	0.004	1.64	< 0.001	0.52
4	Variance risk premium	0.054	< 0.003	3.51**	< 0.003	1.21
5	Multi-asset optimization	—	—	—	—	—
6	Equal risk contribution	—	—	—	—	—
7	Bitcoin volatility	0.178	—	—	—	—
8	Yield curve slope	< 0.08	0.009	2.51	< 0.001	0.84
9	Google Trends fear	0.271	0.038	7.92***	< 0.001	0.67
10	Overnight VIX	0.030	0.005	5.07***	0.003	1.12
11	Calendar anomaly	-0.007	0.012	-2.39*	0.000	0.15
Harvey (2016) threshold						> 3.0

Notes: Partial $r|$VIX is the partial correlation between the signal and 22-day forward realized volatility, controlling for VIX level. IS ΔR^2 is the in-sample increase in R^2 from adding the signal to a VIX-only regression. IS t-stat tests $H_0 : \beta_{\text{signal}} = 0$. OOS R^2 is defined relative to VIX-only forecasts (Equation 6). DM $|t|$ is the absolute Diebold-Mariano statistic with Newey-West standard errors. Families 5–6 are portfolio strategies without direct volatility forecasting components. Family 7 (Bitcoin) is tested for Granger causality rather than regression. $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$. All signals lagged one trading day. Harvey threshold $|t| > 3.0$ required for significance.

Several patterns are noteworthy. First, there is a stark disconnect between in-sample and out-of-sample performance. Google Trends fear (Family 9) achieves a partial correlation of 0.271 and an in-sample ΔR^2 of 0.038 with a highly significant t-statistic of 7.92, yet delivers essentially zero out-of-sample improvement (DM $|t| = 0.67$). This is consistent with overfitting—the fear signal captures contemporaneous co-movement with VIX but has no genuine predictive lead.

Second, the variance risk premium (Family 4) illustrates the distinction between return prediction and volatility prediction. Bollerslev et al. (2009) showed that VRP predicts equity *returns* ($t = 3.51$ in our in-sample test, consistent with their finding). But VRP does not predict *volatility* beyond VIX—unsurprisingly, since VRP is defined as VIX minus realized volatility, making it mechanically correlated with VIX.

Third, the overnight VIX change (Family 10) shows the most promising incremental R^2 (+0.45%), but the improvement vanishes after transaction costs. The overnight signal generates high-frequency weight changes that are too expensive to exploit.

7.2 Strategy Comparison: No Signal-Based Strategy Beats Benchmarks

Table 3 compares the trading strategies derived from each signal family against the 12/VIX and 50/50 SPY/GLD benchmarks over the common evaluation period.

Table 3: Volatility-Timing Strategies: Signal Families vs. Benchmarks

#	Strategy	Sharpe	ΔSharpe	MDD (%)	Turnover	DM $\|t\|$
Benchmarks						
—	Buy-and-hold 50/50 SPY/GLD	0.947	—	−32.5	0.0	—
—	12/VIX (SPY/GLD)	0.870	−0.077	−32.3	8.9×	—
Signal-based strategies						
1	Cross-asset composite	0.810	−0.060	−34.1	11.2×	0.93
2	VIX term structure (contango boost)	0.880	+0.010	−31.8	9.4×	0.31
3	Behavioral sentiment (fear hedge)	0.900	+0.030	−30.2	10.1×	0.72
4	VRP timing	0.782	−0.088	−30.8	12.3×	0.65
4	VRP percentile	0.800	−0.070	−30.5	11.8×	0.58
6	ERC (2-asset)	0.795	−0.054	−13.3	4.2×	1.02
8	Yield curve + VIX combo	0.746	−0.124	−35.1	9.8×	1.15
9	Google Trends fear	0.834	−0.036	−33.7	10.5×	0.48
10	Overnight VIX timing	0.851	−0.019	−31.9	14.7×	0.37
11	Calendar-only	0.658	−0.212	−48.4	2.0×	1.89

Notes: Sharpe ratios computed over the longest common sample for each pair. ΔSharpe is relative to 12/VIX. MDD is maximum drawdown. Turnover is annualized two-way. DM $|t|$ tests whether the signal-based strategy significantly outperforms 12/VIX. All strategies use 5 bps per leg transaction costs and signal.shift(1) lag. Harvey threshold $|t| > 3.0$ required; none passes.

The most striking result is that the static 50/50 SPY/GLD allocation (Sharpe = 0.947) outperforms *every* dynamic strategy, including the 12/VIX benchmark (Sharpe = 0.870). This confirms our earlier finding that VT strategies sacrifice return in exchange for drawdown protection.

Among signal-based strategies, the VIX term structure "contango boost" variant comes closest to matching 12/VIX, with a Sharpe improvement of +0.010, but this is far from statistical significance (DM $|t|$ = 0.31). The behavioral sentiment strategy achieves Sharpe 0.900 but does not significantly outperform 12/VIX (DM $|t| = 0.72$).

7.3 Multi-Asset Portfolio Optimization

Table 4 presents the multi-asset portfolio optimization results from our systematic evaluation of four methods across seven asset subsets.

Table 4: Multi-Asset Portfolio Optimization: 50/50 SPY/GLD vs. Alternatives

Method	N_{assets}	Sharpe	MDD (%)	Beats 50/50?
50/50 SPY/GLD (baseline)	2	1.849	−13.3	—
Equal Weight	3	1.492	−11.2	No
Equal Weight	9	1.285	−8.3	No
Min Variance	3	1.412	−10.8	No
Min Variance	9	1.308	−5.7	No
Max Diversification	3	1.385	−10.5	No
Risk Parity	3	1.401	−10.9	No
Inverse Volatility	2	1.795	−13.3	No
ERC	2	1.795	−13.3	No

Notes: Common evaluation period 2023-01-04 to 2026-03-27 (811 trading days). All methods use monthly hold-and-drift rebalancing, 5 bps per leg TX costs, signal.shift(1). 9-asset universe: SPY, GLD, TLT, QQQ, EEM, VWO, BND, DBC, IEFA. More assets reduce MDD but also reduce Sharpe. The 2-asset ERC and Inverse Volatility converge to approximately 50/50 because SPY vol (19.3%) ≈ GLD vol (18.3%).

The result is unambiguous: no multi-asset strategy beats the simple 50/50 SPY/GLD allocation on Sharpe ratio. Adding assets reduces maximum drawdown (9-asset minimum variance achieves MDD of only −5.7% versus −13.3% for 50/50) but at the cost of lower returns, resulting in lower Sharpe ratios. This is consistent with DeMiguel et al. (2009), who show that estimation error in optimized portfolios offsets diversification benefits.

7.4 Era Stability of VIX Sufficiency

Table 5 presents the VIX forecasting power and strategy performance across five non-overlapping eras.

Table 5: VIX Sufficiency Across Five Market Eras (1993–2026)

Era	Period	N	VIX–RV R^2	β	t-stat	Mean VIX	Mean RV
1. Dot-Com	1993–2000	1,811	0.525	0.810	44.7	18.5	14.6
2. Post-Dot-Com	2000–2007	1,820	0.645	0.868	57.4	19.3	15.7
3. GFC	2007–2012	1,261	0.508	0.957	36.1	26.3	22.4
4. Low-Vol QE	2012–2020	1,908	0.244	0.692	24.8	15.0	11.8
5. COVID/Inflation	2020–2026	1,525	0.309	0.810	26.1	21.0	17.2
Full Sample	1993–2026	8,325	0.514	0.879	93.8	19.5	15.8
Cross-era statistics:			Mean $R^2 = 0.446$		CV $= 0.33$ (time-invariant)		

Notes: R^2 from regressing 22-day forward realized volatility on VIX level. β is the slope coefficient. All t-statistics are highly significant ($p \approx 0$). The lowest R^2 (0.244) occurs during the QE era when VIX was compressed near 15 with low variation; the highest (0.645) occurs during the post-dot-com era with wider VIX range. CV = coefficient of variation of R^2 across eras; CV < 0.50 indicates time-invariance.

The VIX–RV relationship is remarkably stable. The coefficient of variation of R^2 across eras is 0.33—well below the 0.50 threshold we adopt for time-invariance. Importantly, the low R^2 in the QE era (0.244) does not indicate VIX failure; rather, it reflects the mechanical compression of both VIX and RV during this period (VIX stuck near 15, RV near 12), which reduces regression R^2 without impairing VIX's relative forecasting accuracy.

7.5 Competing Signals by Era

Table 6 presents the incremental R^2 of three representative competing signals across the five eras.

Table 6: Incremental R^2 of Competing Signals Across Eras

Signal	Era 1	Era 2	Era 3	Era 4	Era 5	Harvey Pass?
Overnight VIX (abs)	0.0002	0.0001	0.0004	0.0001	0.0003	0/5
VRP proxy	0.0005	0.0002	0.0008	0.0003	0.0004	0/5
Vol momentum 20/60	0.0001	0.0001	0.0006	0.0002	0.0002	0/5

Notes: Incremental R^2 from adding the signal to a VIX-only regression of 22-day forward realized volatility. No signal passes the Harvey (2016) $|t| > 3.0$ threshold in any era. Values are consistently near zero across all regimes, reinforcing VIX sufficiency.

The pattern is striking: incremental R^2 values are uniformly tiny (all below 0.001) across all eras and all signals. VIX sufficiency is not an artifact of any particular market regime.

7.6 12/VIX Strategy by Era

Table 7 shows the 12/VIX strategy performance across eras.

Table 7: 12/VIX Strategy vs. 50/50 Baseline Across Eras

Era	VT Sharpe	BH Sharpe	ΔSharpe	VT MDD	BH MDD	Winner
1. Dot-Com[a]	1.236	1.261	−0.026	−9.2%	−9.9%	50/50
2. Post-Dot-Com[a]	0.128	0.179	−0.051	−31.3%	−26.3%	50/50
3. GFC	0.537	0.669	−0.132	−32.3%	−32.5%	50/50
4. Low-Vol QE	1.116	0.797	+0.319	−12.8%	−13.1%	12/VIX
5. COVID/Inflation	0.963	1.139	−0.176	−20.2%	−20.3%	50/50
12/VIX wins:			1 of 5 eras			

Notes: [a]GLD not available; complement asset is cash (risk-free rate). All strategies use signal.shift(1) and 5 bps per leg TX costs. 12/VIX wins only in the low-volatility QE era when persistent low VIX enables consistent high equity exposure (mean weight ≈ 80%).

The 12/VIX strategy wins in only 1 of 5 eras (the low-volatility QE period). This further reinforces that VT strategies do not generate alpha in the traditional sense; their value lies in drawdown insurance, which is most apparent during high-volatility regimes where VT reduces exposure.

8 Why the Null is Informative

8.1 The Information Aggregation Explanation

Our null result is not a failure to find signal but a positive finding about market efficiency. The VIX option market is one of the deepest and most liquid derivatives markets in the world, with average daily trading volume exceeding 500,000 contracts. This liquidity ensures rapid price discovery, meaning that any signal with predictive content for short-term volatility is quickly impounded into option prices and hence into VIX.

We can formalize this argument. Let Ω_t denote the information set of the representative option trader at time t. Under semi-strong form efficiency of the options market:

$$\mathbb{E}[\sigma^2_{t+1:t+22}|\Omega_t] = f(\mathrm{VIX}_t), \tag{11}$$

where $f(\cdot)$ is monotonically increasing. Any signal $s_t \in \Omega_t$ satisfies:

$$\mathbb{E}[\sigma^2_{t+1:t+22}|\mathrm{VIX}_t, s_t] = \mathbb{E}[\sigma^2_{t+1:t+22}|\mathrm{VIX}_t], \tag{12}$$

i.e., VIX is a sufficient statistic for all information already incorporated in option prices. Our eleven signal families are all elements of Ω_t, which explains the zero incremental forecasting power.

8.2 What Might Break VIX Sufficiency?

The sufficiency argument identifies two conditions under which alternative signals could add value:

1. **Higher-frequency information**: Intraday 5-minute realized volatility *may* capture microstructure dynamics (e.g., opening auctions, lunch-hour liquidity, closing imbalances) that closing-price VIX does not fully reflect. The direct evidence from our pilot study ($N = 37$), however, runs in the opposite direction: daily HAR-ABS (QLIKE = 0.077) outperformed 5-minute HAR-RV (QLIKE = 0.109) by 41.8% in this small sample. A larger-scale replication (planned) is required to determine whether the small-sample direction reverses at scale; until then, we regard the intraday frontier as theoretically plausible but empirically unconfirmed.

2. **Genuinely exogenous shocks**: Events that arrive between option-market pricing updates—geopolitical shocks, pandemic declarations, sudden policy announcements—may create temporary information asymmetries. However, these are by definition unpredictable ex ante.

8.3 The Drawdown Insurance Framework

If VT cannot generate alpha, what is its economic function? We propose that VT is best understood as drawdown insurance, analogous to buying a put option on the portfolio.

Proposition 1 (Drawdown Insurance Cost-Benefit)**.** *For a CRRA investor with risk aversion γ, the certainty-equivalent return of a VT portfolio exceeds that of buy-and-hold when:*

$$\gamma \geq \gamma^* = \frac{\Delta\mu}{\frac{1}{2}(\sigma^2_{BH} - \sigma^2_{VT})}, \tag{13}$$

where $\Delta\mu = \mu_{BH} - \mu_{VT}$ is the return drag from VT, and $\sigma^2_{BH} - \sigma^2_{VT}$ is the variance reduction.

Using our empirical estimates (Table 8), the 12/VIX strategy has a break-even risk aversion of $\gamma^* = 4.5$, and the EWMA VT strategy has $\gamma^* = 4.4$.

Table 8: VT as Drawdown Insurance: Cost-Benefit Analysis

VT Method	Drag	MDD Red.	Cost/1pp	γ^*	Welfare ($\gamma=5$)
12/VIX (SPY/GLD)	3.49%/yr	−8.2 pp	0.43%	4.5	+0.12%/yr
EWMA VT (SPY/GLD)	2.12%/yr	−6.1 pp	0.35%	4.4	+0.15%/yr

Notes: Return drag is the annualized difference in mean return between buy-and-hold and VT strategy. MDD reduction is the decrease in maximum drawdown. Cost per 1 pp MDD is the return drag divided by MDD improvement (in percentage points). Break-even γ is the minimum CRRA risk aversion at which the certainty-equivalent return of VT exceeds buy-and-hold. Welfare gain at $\gamma = 5$ is the annualized certainty-equivalent improvement. Cross-asset validation: 5 assets (SPY, GLD, QQQ, EEM, 0050.TW), 2007–2026.

The insurance framework has a clear practical implication: VT is rational for investors who are moderately to highly risk-averse ($\gamma \geq 4.5$, which encompasses most retail investors and many institutional mandates) but not for risk-neutral or mildly risk-averse investors who should prefer buy-and-hold.

8.4 The Simplicity Premium

A natural question is whether more complex strategies justify their complexity. Our meta-analysis of 14 live-tracked strategies reveals a *simplicity premium*: the Spearman correlation between the number of estimated parameters and the Sharpe ratio is $\rho = 0.077$ ($p = 0.794$), indicating zero relationship between complexity and performance. The best-performing strategies (50/50 SPY/GLD, 12/VIX, Piecewise VT) have zero or one estimated parameter.

This echoes DeMiguel et al.'s (2009) finding that $1/N$ portfolios outperform optimized alternatives. In the VT context, the simplicity premium arises because:

1. Simple rules have fewer parameters to estimate, reducing overfitting.

2. Smooth weight functions (like 12/VIX) produce low turnover, saving transaction

costs.

3. Zero-parameter rules are immune to estimation window and specification choices.

9 Conclusion

We set out to answer a straightforward question: can any observable signal beat VIX for equity volatility forecasting and volatility timing? After a systematic evaluation of eleven pre-specified signal families across 33 years of data, five market eras, and a unified pipeline with strict lag enforcement, transaction-cost accounting, and Harvey (2016) multiple-testing control, the answer is no.

This null is not a failure but an informative finding. It reflects the efficiency of the options market, which aggregates heterogeneous information—cross-asset correlations, macroeconomic indicators, sentiment, term structure dynamics—into a single summary statistic. Each of the eleven signal families we tested represents one channel of information that option traders already incorporate into their pricing. VIX, as the price-weighted consensus, already contains them.

The practical implications are clear:

1. **For researchers**: The daily-frequency, closing-price frontier is exhausted. Gains require genuinely exogenous information outside the option-market information set; the 5-minute realized-measure frontier is theoretically plausible but is not yet empirically confirmed (our pilot study with $N = 37$ found daily HAR-ABS *outperformed* 5-minute HAR-RV by 41.8%, a result that should be tested at scale before positing 5-minute data as a reliable improvement channel).

2. **For practitioners**: The simplest VIX-based rule (12/VIX) cannot be improved by adding signals. Complexity reduces returns through estimation error, higher turnover, and transaction costs. The optimal allocation for most investors is either buy-and-hold 50/50 SPY/GLD (if $\gamma < 4.5$) or 12/VIX VT (if $\gamma \geq 4.5$).

3. **For regulators and risk managers**: VIX sufficiency is time-invariant (CV =

0.33 across 33 years). Models based on VIX do not need frequent recalibration, and regulatory frameworks (e.g., Basel III/IV) that reference VIX for market risk are on solid empirical ground.

We conclude with a broader observation. In financial econometrics, null results are underreported relative to their information content. Our systematic horse race, with its pre-specified signal families and stringent statistical controls, provides stronger evidence than any individual positive finding could. The 36 null results accumulated across this research program are not 36 failures; they are 36 independent confirmations that the option market efficiently aggregates volatility-relevant information.

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